

Arithmetic in Spark `calcDivB`

Source: `source/physics/Hydro/HydroMain/Spark/Hydro_funcs.F90`, routine `calcDivB` (lines 573–620). This document transcribes the magnetic-divergence diagnostic exactly as implemented, including its dimensional preprocessor conditions.

1 Notation and execution domain

Let $q = (i, j, k)$ denote an interior cell in `blockDesc%limits`. Let $B_{x,q}$, $B_{y,q}$, and $B_{z,q}$ be the cell-centered values in `MAGX_VAR`, `MAGY_VAR`, and `MAGZ_VAR`. The block spacings returned by `blockDesc%deltas` are

$$h_x = \text{del}(\text{IAXIS}), \quad h_y = \text{del}(\text{JAXIS}), \quad h_z = \text{del}(\text{KAXIS}). \quad (1)$$

The output is $D_{B,q} = U(\text{DIVB}, q)$, corresponding to `solnData(DIVB_VAR,i,j,k)`. The computational body is enabled only when `SPARK_GLM` is defined. Without that definition, this routine does not update the output field.

2 Centered-difference arithmetic

At the start of each interior-cell iteration, the scalar accumulator is

$$\text{divB} \leftarrow 0. \quad (2)$$

When $N = \text{NDIM} > 1$, the code assigns the x contribution and then adds the y contribution:

$$(\partial_x B_x)_q^{\text{FD}} = \frac{B_{x,i+1,j,k} - B_{x,i-1,j,k}}{2h_x}, \quad (3)$$

$$(\partial_y B_y)_q^{\text{FD}} = \frac{B_{y,i,j+1,k} - B_{y,i,j-1,k}}{2h_y}. \quad (4)$$

When $N = 3$, it additionally adds

$$(\partial_z B_z)_q^{\text{FD}} = \frac{B_{z,i,j,k+1} - B_{z,i,j,k-1}}{2h_z}. \quad (5)$$

Finally, the accumulator is stored in `DIVB_VAR`.

3 Exact dimension-dependent output

For a build with SPARK_GLM, the complete output formula is

$$D_{B,q} = \begin{cases} 0, & N = 1, \\ \frac{B_{x,i+1,j,k} - B_{x,i-1,j,k}}{2h_x} + \frac{B_{y,i,j+1,k} - B_{y,i,j-1,k}}{2h_y}, & N = 2, \\ \frac{B_{x,i+1,j,k} - B_{x,i-1,j,k}}{2h_x} + \frac{B_{y,i,j+1,k} - B_{y,i,j-1,k}}{2h_y} + \frac{B_{z,i,j,k+1} - B_{z,i,j,k-1}}{2h_z}, & N = 3. \end{cases} \quad (6)$$

One-dimensional behavior: the x derivative is inside `#if NDIM>1`, together with the y derivative. Therefore the one-dimensional implementation writes zero, even if B_x varies with x . This records the source behavior; it does not substitute the usual one-dimensional derivative.

4 Interpretation and stencil requirements

For two and three dimensions on a uniform Cartesian grid, these formulas are the standard centered approximation to

$$\nabla \cdot \mathbf{B} = \partial_x B_x + \partial_y B_y + \partial_z B_z, \quad (7)$$

with the inactive directional derivatives omitted. For smooth fields, the directional truncation errors are of order h_d^2 , for example

$$\frac{B_x(x + h_x) - B_x(x - h_x)}{2h_x} = \partial_x B_x(x) + \frac{h_x^2}{6} \partial_x^3 B_x(x) + O(h_x^4). \quad (8)$$

This last equation explains the stencil accuracy; it is not additional arithmetic executed by the routine.

The code uses block spacings directly, with no face-area/volume weighting or cylindrical/spherical metric terms. It takes signed derivatives and does not apply an absolute value, normalization, limiter, or floor. The output has magnetic-field units divided by length. Only interior output cells are assigned; output guard cells retain their incoming values. Neighbor reads require one available cell beyond each interior boundary in every differentiated direction, normally supplied by guard cells. The routine computes a diagnostic and does not modify the magnetic field or perform divergence cleaning.